Customer Stories / Government / United States

2023

Bringing GIS to the AWS Cloud at ATL, the World's Busiest Airport

Learn how Hartsfield-Jackson Atlanta International Airport migrated to the cloud and increased agility, scalability, and innovation in managing its complex operations.

Overview | Opportunity | Solution | Outcome | AWS Services Used

1 week

to migrate to AWS

99.9%

uptime for Esri's ArcGIS

Deploys

applications in hours instead of days

Scales

dynamically without advance planning

Reduced

Overview

There's no such thing as a simple airport. That's especially true in Atlanta, where Hartsfield-Jackson Atlanta International Airport (ATL)—with five runways and 4,700 acres of land—manages to be the world's busiest airport as measured by foot traffic, with 75.7 million passengers in 2021. It also ranks among the most efficient airports, according to the Air Transport Research Society's productive and efficiency metrics.

With millions of moving pieces, airports don't have much room for error. That's why they typically rely on geographic information systems (GIS) to help them manage infrastructure such as utilities, terminals, parking garages, and taxiways. However, until 2016, ATL's underlying IT architecture just couldn't meet the needs of a successful and efficient GIS solution: slow, expensive, and complicated to manage.

ATL wanted a cost-effective, elastic, and agile GIS solution, so it turned to Axim Geospatial (Axim), an Amazon Web Services (AWS) Partner. As experts in Esri's ArcGIS software, one of the world's leading GIS solutions, Axim could provide ATL with guidance and a detailed migration map from on premises to the cloud; as an AWS Partner, it knew how the cloud could help ATL improve operations while saving time and money. After plans were in place, the migration was complete within a week, and ATL reduced technical debt and increased efficiency, productivity, agility, and scalability.

Opportunity | Using AWS to Improve Operations

Team members at ATL used ArcGIS desktop solutions proficiently, but as the airport's workloads grew, they noticed that the on-premises server infrastructure couldn't keep up. Creating an application took days, if it was feasible at all. The airport paid huge upfront costs for the biggest available servers, all of which had to last through multiyear procurement cycles; receiving, implementing, configuring, and ramping up those servers would take 6 months. With the airport's technical staff stretched to address higher priority needs, ATL often lost in-house IT GIS experts to the private sector before server implementation and application development could be complete

In 2013, ATL began using ArcGIS Online, Esri's software as a service, which proved to the airport that web-based services were viable. When ATL decided to migrate in 2015, it reached out to Axim, with which it had a decade-long relationship. Axim performed rigorous assessments of ATL's needs and capability requirements and created plans for the cloud migration, detailing everything from the number of servers to the outcomes of different architectural layouts. "At the time, other options weren't where AWS was in terms of the ability to log in and spin up a server," says Brian Haren, GIS program manager at ATL. "The hosting environment on AWS was the best fit for what we needed to do."

Solution | Using Amazon EC2 and Amazon S3 to Scale Dynamically and Cost Effectively

It took a week to get ATL onto AWS in 2016. "There were no challenges," says Mr. Haren. "Axim team members scoped everything out ahead of time. They purchased what we needed in the AWS environment. We simply handed them our licensing, and they stood up our Esri software products."

AWS works with leading transportation technologies to build cloud solutions for airports, seaports, departments of transportation, and more. On AWS, the airport reduced technical debt, dramatically cutting the costs and headaches associated with on-premises servers. "Hartsfield-Jackson Atlanta International Airport has always been a leader in the adoption of GIS technology to help it manage its complex facilities," says Terry Bills, transportation industry director at Esri. "So it was no surprise that it turned to AWS to get to the next level, and it points the way for other airports to follow."

ATL runs ArcGIS on Amazon Elastic Compute Cloud (Amazon EC2), a broad, deep, compute service that has more than 500 instances and multiple processors, storage solutions, networking options, operating systems, and purchase models from which to choose. The airport's extensive datasets are stored in Amazon Simple Storage Service (Amazon S3), an object storage service offering industry-leading scalability, data availability, security, and performance. "On AWS, we could start doing the work that we had wanted to do years ago but hadn't been able to," says Mr. Haren. "It was almost an overnight shift from printing out maps on paper or plotters to web-services delivery." After ATL's ArcGIS was on AWS, its uptime increased to 99.9 percent.

On AWS, ATL also increased efficiency and staff productivity. Employees use a cloud-based mobile application to access ArcGIS in the field: the maintenance team uses it to track tasks through web maps, and the airport operations group uses it to manage parking lots, runways, and taxiways. “If our users get us their requirements, we can have a web map up and operating for them within hours,” says Mr. Haren. “We can make that promise only because we run on AWS.” Plus, staff like Mr. Haren no longer have to spend time leading technical projects.

ATL relies on Axim for that, freeing its employees to focus on work that only they can do. “Moving onpremises enterprise GIS to cloud environments has been an IT and geospatial industry trend for more than a decade, and for ATL, it made perfect sense to use the AWS Cloud,” says Kevin Stewart, vice president of sales at Axim. “The shared vision and our team of cloud experts made the transition seamless.”

The airport also began using Amazon WorkSpaces, a fully managed desktop virtualization service for Windows and Linux that can access resources from any supported device. “When our folks come in and log in to work, they’re logging into the remote desktop in Amazon WorkSpaces,” says Mr. Haren. During theCOVID-19 pandemic, ATL staff could use Amazon WorkSpaces immediately to work from home.

The pay-as-you-go scaling of Amazon EC2 instances also helps ATL dramatically reduce costs and increase innovation—both crucial for developing new tools. “We might take 3–6 months to develop, implement, and test a new product,” says David Wright, assistant director of GIS at ATL. “We can start on a small Amazon EC2 instance and not have to pay high dollar for something that won’t deploy for a while.” When projects are ready, ATL uses Amazon EC2 to scale them dynamically, without having to secure capacity ahead of time. “If there are hiccups with a project, we can delay deployment,” says Mr. Haren. “We can do just-in-time launches and wait until the last minute to pick exactly what we need in the AWS environment.”

Outcome | Increasing Agility and Innovation on AWS

Since migrating to AWS, ATL has reduced its technical debt, increased staff productivity, and become more agile and innovative, which helps it maintain its status as one of the world’s most efficient airports. “When migrating from a traditional server to the cloud, the cost savings over time will significantly outweigh transportation groups’ fears about not having everything on premises and supported by their own IT group,” says Wright. “Not having to spend months and years gearing up for something that we can do in a relatively short amount of time is a tremendous return on investment that can’t be monetized.”

About Hartsfield-Jackson Atlanta International Airport

Hartsfield-Jackson Atlanta International Airport offers nonstop flights to more than 150 domestic and 70 international destinations as well as 300 commercial venues. It was the first airport to serve more than 100 million passengers in 1 year.

AWS Services Used

Amazon Elastic Compute Cloud (Amazon EC2)

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

Learn more »

Amazon Simple Storage Service (Amazon S3)

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

Learn more »

Amazon WorkSpaces

Amazon WorkSpaces is a fully managed desktop virtualization service for Windows, Linux, and Ubuntu that allows you to access resources from any supported device, including Amazon WorkSpaces Thin Client.

Learn more »

More Government Customer Stories

MALAYSIA

SMART SELANGOR

Smart Selangor Pioneers a Connected Future with…

SINGAPORE

GOVTECH SINGAPORE

GovTech Collaborates with AWS Generative AI…

This case study explores how GovTech,

2020

2024

AUSTRALIA

Tasmanian Government

Department of Premier and Cabinet Tasmania…

Discover how the central agency of the

2024

JAPAN

Tsukuba City

Implementing Blockchain and Personal ID-based Online Voting in Tsukuba's Society 5.0 Social Implementation Trials

2020

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

Contact Sales

2

What Is a Data Lake?
What is Artificial Intelligence (AI)?
What is Generative AI?
What is Machine Learning (ML)?
AWS Cloud Security
What's New
Blogs
Press Releases

Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language
عربي |
Bahasa Indonesia |
Deutsch |
English |
Español |
Français |
Italiano |
Português |
Tiếng Việt |
Türkçe |
Русский |
ไทย |
日本語 |
한국어 |
中文 (简体) |
中文 (繁體)

Privacy
|
Accessibility
|
Site Terms
|
Cookie Preferences
|